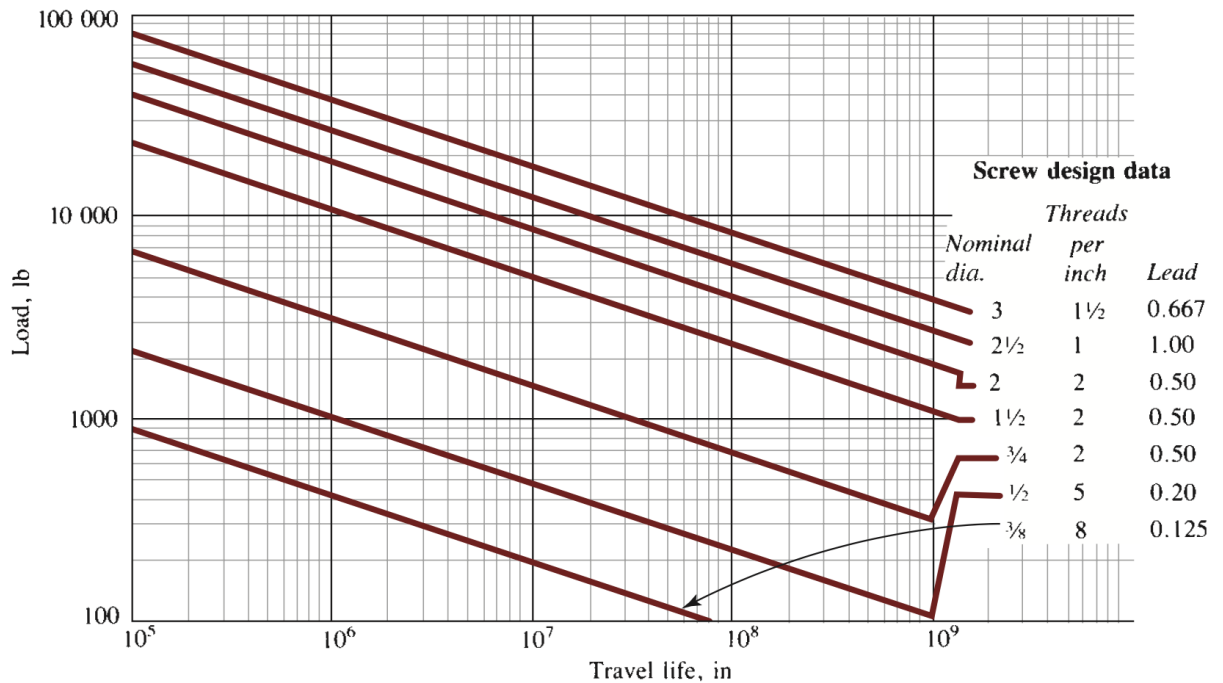




3. Compute the torque (in lb · in) required to drive the screw using the following equation:

$$T_u = 0.177FL$$

where F is the applied load in lbs,

L is the lead (from the graph above) in inches

4. Compute the power required (in hp) given a travel speed:

- (a) Calculate the rotational speed required (in rpm):

$$n = (V) \left(\frac{1 \text{ rev}}{L} \right) \left(\frac{60 \text{ sec}}{\text{min}} \right)$$

where V is the travel speed in in/s,

L is the lead in inches

- (b) The power required (in hp) is the following:

$$P = \frac{Tn}{63000}$$

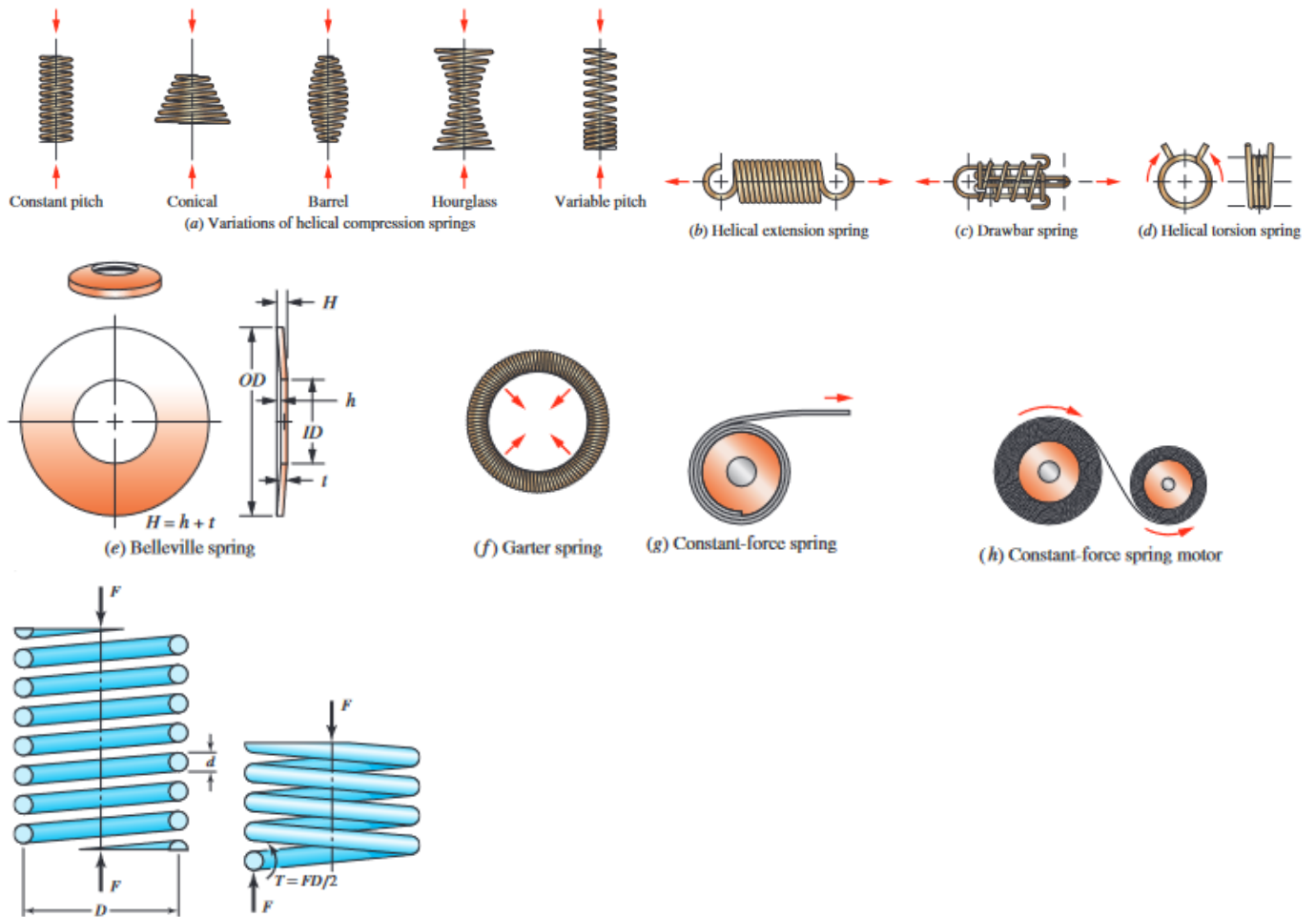
where T is the required torque in lb · in,

n is the rotational speed in rpm

5.3 Springs

This section is best for Static Helical Springs, Chapter 10 of shigley has pretty good depth on all spring types if we get something else

5.3.1 Anatomy



D = Mean Diameter (Distance to centre of wire)

d = Wire Diameter (Distance to centre of wire)

Outer Diameter = $OD = D + d$

Inner Diameter = $ID = D - d$

Max Shear Stress = $\tau_{max} = \frac{T_r}{J} + \frac{F}{A}$

This one is the above one with all subs built in: $\tau = \frac{8FD}{\pi d^3} + \frac{4F}{\pi d^2}$

Spring index (best between 4 and 12) = $C = \frac{D}{d}$

Changing that bitch again: $\tau = K_s \frac{8FD}{\pi d^3}$

How could you forget K_s is stress-correction factor: $K_s = \frac{2C+1}{2C}$

This shit is driving me up the Wahl factor: $K_W = \frac{4C-1}{4C-4} + \frac{0.615}{C}$

But of course we prefer Bergsträsser Factor: $K_B = \frac{4C+2}{4C-3}$

Why do we need to account for Curvature factor? Its supposed to be curved: $K_C = \frac{K_B}{K_s} = \frac{2C(4C+2)}{(4C-3)(2C+1)}$

Most of those don't matter just use: $\tau = K_B \frac{8FD}{\pi d^3}$

Oh shit we got these using Castigliano's, wish I paid more attention in 360

N = Number of coils

N_a = Number of active coils

Ignore this one apparently: $U = \frac{T^2 l}{2GJ} + \frac{F^2 l}{2AG}$

This one for sure though (also $N = N_a$ idk why): $U = \frac{4F^2 D^3 N}{d^4 G} + \frac{2F^2 DN}{d^2 G}$

Total deflection: $= \frac{\partial U}{\partial F} = \frac{8FD^3 N}{d^4 G} + \frac{4FDN}{d^2 G} = \frac{8FD^3 N}{d^4 G} \left(1 + \frac{1}{2C^2}\right) \approx \frac{8FD^3 N}{d^4 G}$

Spring constant: $k = \frac{F}{y} \approx \frac{d^4 G}{8D^3 N}$

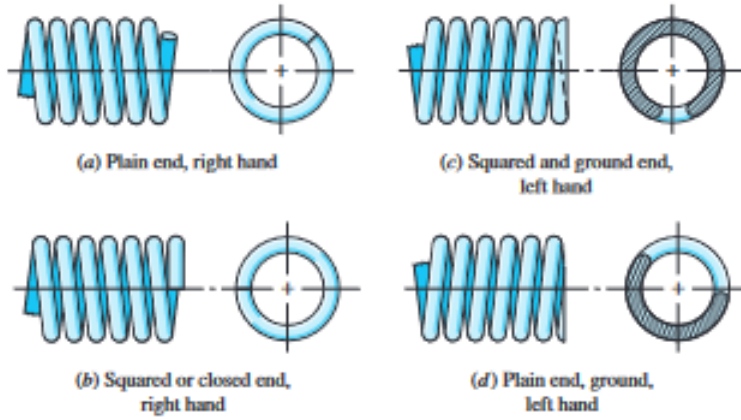


Table 10–1 Formulas for the Dimensional Characteristics of Compression Springs (N_a = Number of Active Coils)

Term	Type of Spring Ends			
	Plain	Plain and Ground	Squared or Closed	Squared and Ground
End coils, N_e	0	1	2	2
Total coils, N_t	N_a	$N_a + 1$	$N_a + 2$	$N_a + 2$
Free length, L_0	$pN_a + d$	$p(N_a + 1)$	$pN_a + 3d$	$pN_a + 2d$
Solid length, L_s	$d(N_t + 1)$	dN_t	$d(N_t + 1)$	dN_t
Pitch, p	$(L_0 - d)/N_a$	$L_0/(N_a + 1)$	$(L_0 - 3d)/N_a$	$(L_0 - 2d)/N_a$

Critical deflection: $y_{cr} = L_0 C_1' \left[1 - \left(1 - \frac{C_2'}{\lambda_{eff}^2}\right)^{1/2}\right]$

Table 10–2 End-Condition Constants α for Helical Compression Springs*

End Condition	Constant α
Spring supported between flat parallel surfaces (fixed ends)	0.5
One end supported by flat surface perpendicular to spring axis (fixed); other end pivoted (hinged)	0.707
Both ends pivoted (hinged)	1
One end clamped; other end free	2

*Ends supported by flat surfaces must be squared and ground.

Effective slenderness ratio: $\lambda_{eff} = \frac{\alpha L_0}{D}$

Book really said "random elastic constants": $C'_1 = \frac{E}{2(E-G)}$, $C'_2 = \frac{2\pi^2(E-G)}{2G+E}$

Absolute stability occurs if: $L_0 < \frac{\pi D}{\alpha} [\frac{2(E-G)}{2G+E}]^{1/2}$, But for steels: $L_0 < 2.63 \frac{D}{\alpha}$

Table 10–3 High-Carbon and Alloy Spring Steels

Name of Material	Similar Specifications	Description
Music wire, 0.80–0.95C	UNS G10850 AISI 1085 ASTM A228-51	This is the best, toughest, and most widely used of all spring materials for small springs. It has the highest tensile strength and can withstand higher stresses under repeated loading than any other spring material. Available in diameters 0.12 to 3 mm (0.005 to 0.125 in). Do not use above 120°C (250°F) or at subzero temperatures.
Oil-tempered wire, 0.60–0.70C	UNS G10650 AISI 1065 ASTM 229-41	This general-purpose spring steel is used for many types of coil springs where the cost of music wire is prohibitive and in sizes larger than available in music wire. Not for shock or impact loading. Available in diameters 3 to 12 mm (0.125 to 0.5000 in), but larger and smaller sizes may be obtained. Not for use above 180°C (350°F) or at subzero temperatures.
Hard-drawn wire, 0.60–0.70C	UNS G10660 AISI 1066 ASTM A227-47	This is the cheapest general-purpose spring steel and should be used only where life, accuracy, and deflection are not too important. Available in diameters 0.8 to 12 mm (0.031 to 0.500 in). Not for use above 120°C (250°F) or at subzero temperatures.
Chrome-vanadium	UNS G61500 AISI 6150 ASTM 231-41	This is the most popular alloy spring steel for conditions involving higher stresses than can be used with the high-carbon steels and for use where fatigue resistance and long endurance are needed. Also good for shock and impact loads. Widely used for aircraft-engine valve springs and for temperatures to 220°C (425°F). Available in annealed or pretempered sizes 0.8 to 12 mm (0.031 to 0.500 in) in diameter.
Chrome-silicon	UNS G92540 AISI 9254	This alloy is an excellent material for highly stressed springs that require long life and are subjected to shock loading. Rockwell hardnesses of C50 to C53 are quite common, and the material may be used up to 250°C (475°F). Available from 0.8 to 12 mm (0.031 to 0.500 in) in diameter.

Table 10–4 Constants A and m of $S_{ut} = A/d^m$ for Estimating Minimum Tensile Strength of Common Spring Wires

Material	ASTM No.	Exponent m	Diameter, In	A , kpsi · In ^{m}	Diameter, mm	A , MPa · mm ^{m}	Relative Cost of Wire
Music wire*	A228	0.145	0.004–0.256	201	0.10–6.5	2211	2.6
OQ&T wire [†]	A229	0.187	0.020–0.500	147	0.5–12.7	1855	1.3
Hard-drawn wire [‡]	A227	0.190	0.028–0.500	140	0.7–12.7	1783	1.0
Chrome-vanadium wire [§]	A232	0.168	0.032–0.437	169	0.8–11.1	2005	3.1
Chrome-silicon wire	A401	0.108	0.063–0.375	202	1.6–9.5	1974	4.0
302 Stainless wire [#]	A313	0.146	0.013–0.10	169	0.3–2.5	1867	7.6–11
		0.263	0.10–0.20	128	2.5–5	2065	
		0.478	0.20–0.40	90	5–10	2911	
Phosphor-bronze wire**	B159	0	0.004–0.022	145	0.1–0.6	1000	8.0
		0.028	0.022–0.075	121	0.6–2	913	
		0.064	0.075–0.30	110	2–7.5	932	

*Surface is smooth, free of defects, and has a bright, lustrous finish.

[†]Has a slight heat-treating scale which must be removed before plating.

[‡]Surface is smooth and bright with no visible marks.

[§]Aircraft-quality tempered wire, can also be obtained annealed.

^{||}Tempered to Rockwell C49, but may be obtained untempered.

[#]Type 302 stainless steel.

**Temper CA510.

Ultimate Tensile Strength: $S_{ut} = \frac{A}{d^m}$

Torsional yield strength: $S_{sy} = 0.577S_y$ same as $0.35S_{ut} \leq S_{sy} \leq 0.52S_{ut}$

Table 10–5 Mechanical Properties of Some Spring Wires

Material	Elastic Limit, Percent of S_u		Diameter d , in	E		G	
	Tension	Torsion		Mpsi	GPa	Mpsi	GPa
Music wire A228	65–75	45–60	<0.032	29.5	203.4	12.0	82.7
			0.033–0.063	29.0	200	11.85	81.7
			0.064–0.125	28.5	196.5	11.75	81.0
			>0.125	28.0	193	11.6	80.0
HD spring A227	60–70	45–55	<0.032	28.8	198.6	11.7	80.7
			0.033–0.063	28.7	197.9	11.6	80.0
			0.064–0.125	28.6	197.2	11.5	79.3
			>0.125	28.5	196.5	11.4	78.6
Oil tempered A239	85–90	45–50		28.5	196.5	11.2	77.2
Valve spring A230	85–90	50–60		29.5	203.4	11.2	77.2
Chrome-vanadium A231	88–93	65–75		29.5	203.4	11.2	77.2
A232	88–93			29.5	203.4	11.2	77.2
Chrome-silicon A401	85–93	65–75		29.5	203.4	11.2	77.2
Stainless steel							
A313*	65–75	45–55		28	193	10	69.0
17-7PH	75–80	55–60		29.5	208.4	11	75.8
414	65–70	42–55		29	200	11.2	77.2
420	65–75	45–55		29	200	11.2	77.2
431	72–76	50–55		30	206	11.5	79.3
Phosphor-bronze B159	75–80	45–50		15	103.4	6	41.4
Beryllium-copper B197	70	50		17	117.2	6.5	44.8
Inconel alloy X-750	65–70	40–45		31	213.7	11.2	77.2

*Also includes 302, 304, and 316.

Table 10–6 Maximum Allowable Torsional Stresses for Helical Compression Springs in Static Applications

Material	Maximum Percent of Tensile Strength	
	Before Set Removed (includes K_W or K_B)	After Set Removed (includes K_s)
Music wire and cold-drawn carbon steel	45	60–70
Hardened and tempered carbon and low-alloy steel	50	65–75
Austenitic stainless steels	35	55–65
Nonferrous alloys	35	55–65

$$S_{sy} = \tau_{all} = 0.56S_{ut}$$

Table A-28 Decimal Equivalents of Wire and Sheet-Metal Gauges* (All Sizes Are Given in Inches)

Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard†	Manufacturers Standard	Steel Wire or Washburn & Moen	Music Wire	Stubs Steel Wire	Twist Drill
Principal Use:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lbf/in ²	Ferrous Sheet	Ferrous Wire Except Music Wire	Music Wire	Steel Drill Rod	Twist Drills and Drill Steel
7/0			0.500		0.490			
6/0	0.580 0		0.468 75		0.461 5	0.004		
5/0	0.516 5		0.437 5		0.430 5	0.005		
4/0	0.460 0	0.454	0.406 25		0.393 8	0.006		
3/0	0.409 6	0.425	0.375		0.362 5	0.007		
2/0	0.364 8	0.380	0.343 75		0.331 0	0.008		
0	0.324 9	0.340	0.312 5		0.306 5	0.009		
1	0.289 3	0.300	0.281 25		0.283 0	0.010	0.227	0.228 0
2	0.257 6	0.284	0.265 625		0.262 5	0.011	0.219	0.221 0
3	0.229 4	0.259	0.25	0.239 1	0.243 7	0.012	0.212	0.213 0
4	0.204 3	0.238	0.234 375	0.224 2	0.225 3	0.013	0.207	0.209 0
5	0.181 9	0.220	0.218 75	0.209 2	0.207 0	0.014	0.204	0.205 5
6	0.162 0	0.203	0.203 125	0.194 3	0.192 0	0.016	0.201	0.204 0
7	0.144 3	0.180	0.187 5	0.179 3	0.177 0	0.018	0.199	0.201 0
8	0.128 5	0.165	0.171 875	0.164 4	0.162 0	0.020	0.197	0.199 0
9	0.114 4	0.148	0.156 25	0.149 5	0.148 3	0.022	0.194	0.196 0
10	0.101 9	0.134	0.140 625	0.134 5	0.135 0	0.024	0.191	0.193 5
11	0.090 74	0.120	0.125	0.119 6	0.120 5	0.026	0.188	0.191 0
12	0.080 81	0.109	0.109 357	0.104 6	0.105 5	0.029	0.185	0.189 0
13	0.071 96	0.095	0.093 75	0.089 7	0.091 5	0.031	0.182	0.185 0
14	0.064 08	0.083	0.078 125	0.074 7	0.080 0	0.033	0.180	0.182 0
15	0.057 07	0.072	0.070 312 5	0.067 3	0.072 0	0.035	0.178	0.180 0
16	0.050 82	0.065	0.062 5	0.059 8	0.062 5	0.037	0.175	0.177 0
17	0.045 26	0.058	0.056 25	0.053 8	0.054 0	0.039	0.172	0.173 0

Table A-28 Decimal Equivalents of Wire and Sheet-Metal Gauges* (All Sizes Are Given in Inches) (Continued)

Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard [†]	Manu- facturers Standard	Steel Wire or Washburn & Moen	Music Wire	Stubs Steel Wire	Twist Drill
Principal Use:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lb/ft ²	Ferrous Sheet	Ferrous Wire Except Music Wire	Music Wire	Steel Drill Rod	Twist Drills and Drill Steel
18	0.040 30	0.049	0.05	0.047 8	0.047 5	0.041	0.168	0.169 5
19	0.035 89	0.042	0.043 75	0.041 8	0.041 0	0.043	0.164	0.166 0
20	0.031 96	0.035	0.037 5	0.035 9	0.034 8	0.045	0.161	0.161 0
21	0.028 46	0.032	0.034 375	0.032 9	0.031 7	0.047	0.157	0.159 0
22	0.025 35	0.028	0.031 25	0.029 9	0.028 6	0.049	0.155	0.157 0
23	0.022 57	0.025	0.028 125	0.026 9	0.025 8	0.051	0.153	0.154 0
24	0.020 10	0.022	0.025	0.023 9	0.023 0	0.055	0.151	0.152 0
25	0.017 90	0.020	0.021 875	0.020 9	0.020 4	0.059	0.148	0.149 5
26	0.015 94	0.018	0.018 75	0.017 9	0.018 1	0.063	0.146	0.147 0
27	0.014 20	0.016	0.017 187 5	0.016 4	0.017 3	0.067	0.143	0.144 0
28	0.012 64	0.014	0.015 625	0.014 9	0.016 2	0.071	0.139	0.140 5
29	0.011 26	0.013	0.014 062 5	0.013 5	0.015 0	0.075	0.134	0.136 0
30	0.010 03	0.012	0.012 5	0.012 0	0.014 0	0.080	0.127	0.128 5
31	0.008 928	0.010	0.010 937 5	0.010 5	0.013 2	0.085	0.120	0.120 0
32	0.007 950	0.009	0.010 156 25	0.009 7	0.012 8	0.090	0.115	0.116 0
33	0.007 080	0.008	0.009 375	0.009 0	0.011 8	0.095	0.112	0.113 0
34	0.006 305	0.007	0.008 593 75	0.008 2	0.010 4		0.110	0.111 0
35	0.005 615	0.005	0.007 812 5	0.007 5	0.009 5		0.108	0.110 0
36	0.005 000	0.004	0.007 031 25	0.006 7	0.009 0		0.106	0.106 5
37	0.004 453		0.006 640 625	0.006 4	0.008 5		0.103	0.104 0
38	0.003 965		0.006 25	0.006 0	0.008 0		0.101	0.101 5
39	0.003 531				0.007 5		0.099	0.099 5
40	0.003 145				0.007 0		0.097	0.098 0

*Specify sheet, wire, and plate by stating the gauge number, the gauge name, and the decimal equivalent in parentheses.

[†]Reflects present average and weights of sheet steel.

$\frac{\partial^2 u}{\partial x^2} = \frac{W}{kg l^2} \frac{\partial^2 u}{\partial t^2} // g = \text{acceleration from gravity}, l = \text{length of spring between plates}, W = \text{weight of spring}$

Fundamental Frequencies: $\omega = m\pi\sqrt{\frac{kg}{W}}, m = 1, 2, 3, \dots$

Both ends always contact plates: $f = \frac{1}{2}\sqrt{\frac{kg}{W}}$

One end always contacts plate: $f = \frac{1}{4}\sqrt{\frac{kg}{W}}$ (also works if being driven by a sine wave machine)

$W = AL\gamma = \frac{\pi^2 d^2 D N_a \gamma}{4}$ Where γ is specific weight

Always make sure f_{cr} is 15-20 times higher than operating frequency to avoid resonance

I don't understand the next two lines

Unpeened: $S_{sa} = 35\text{kpsi}(241\text{MPa}), S_{sm} = 55\text{kpsi}(379\text{MPa})$

Peened: $S_{sa} = 57.5\text{kpsi}(398\text{MPa}), S_{sm} = 77.5\text{kpsi}(534\text{MPa})$

Goodman Criteria: $S_{se} = \frac{S_{sa}}{1 - \frac{S_{sm}}{S_{su}}}$

Gerber Criteria: $S_{se} = \frac{S_{sa}}{1 - (\frac{S_{sm}}{S_{su}})^2}$

Sines Failure Criterion: $S_{su} = 0.67 S_{ut}$

$F_a = \frac{F_{max} - F_{min}}{2}$

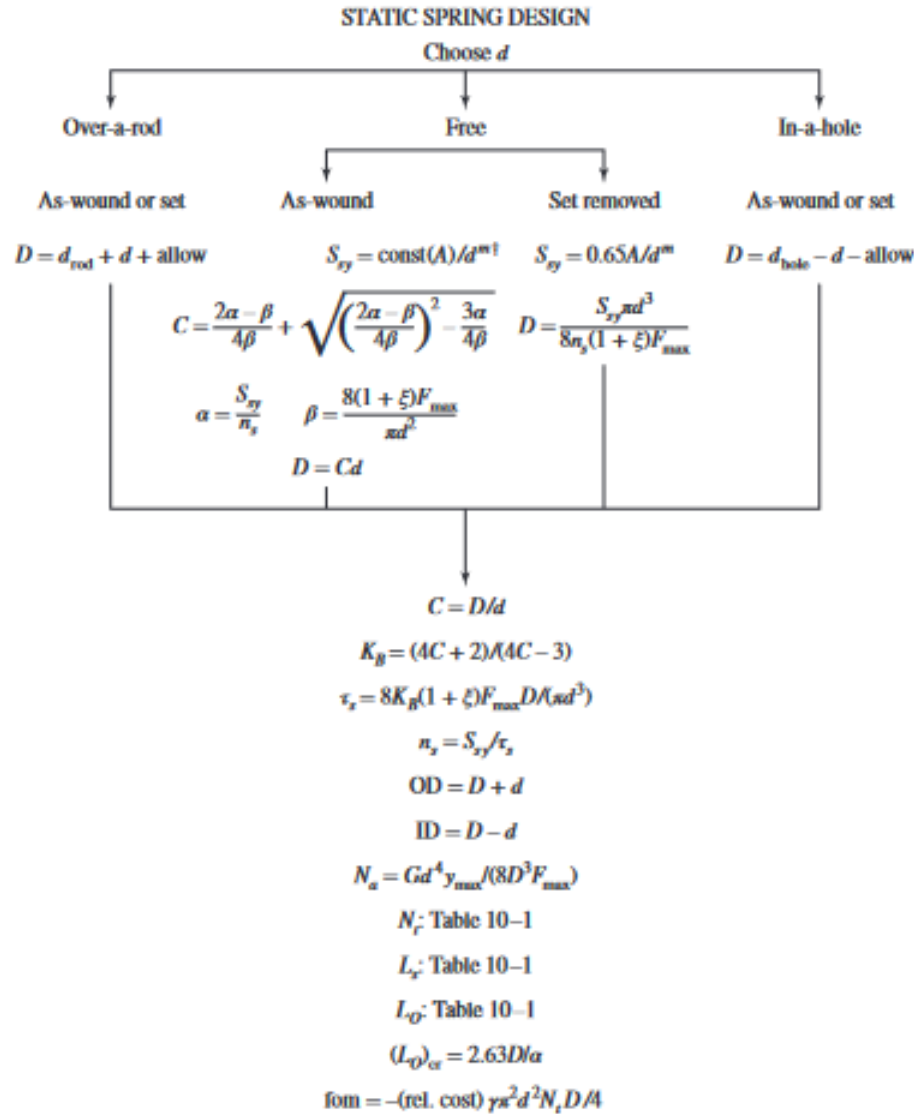
$F_m = \frac{F_{max} + F_{min}}{2}$

$\tau_a = K_B \frac{8 F_a D}{\pi d^3}$

$$\tau_m = K_B \frac{8F_m D}{\pi d^3}$$

5.3.2 Design for Static Service

Design requirements: $4 \leq C \leq 12$, $3 \leq N_a \leq 15$, $\xi \geq 0.15$, $n_s \geq 1.2$ Note n_s is the safety factor at solid height
 fom = $-(\text{relative material cost}) \frac{\gamma \pi^2 d^2 N_t D}{4}$



Print or display: $d, D, C, \text{OD}, \text{ID}, N_a, N_t, L_s, L_Q, (L_Q)_{cr}, n_s, \text{fom}$

Build a table, conduct design assessment by inspection

Eliminate infeasible designs by showing active constraints

Choose among satisfactory designs using the figure of merit

[†]const is found from Table 10-6.

Recall: $\tau = \frac{S_{sy}}{n_s} = K_B \frac{8F_s D}{\pi d^3} = \frac{4C+2}{4C-3} \left[\frac{8(1+\xi)F_{\max} C}{\pi d^2} \right]$

Let: $\alpha = \frac{S_{sy}}{n_s}$ and $\beta = \frac{8(1+\xi)F_{max}}{\pi d^2}$
 $\therefore C = \frac{2\alpha-\beta}{4\beta} + \sqrt{\left(\frac{2\alpha-\beta}{4\beta}\right)^2 - \frac{3\alpha}{4\beta}}$ this gives two answers, always pick the larger

5.3.3 Iterate wire diameter

- Gather needed info:

Based on the wire you need, use these tables to pick an A, m E and G value (you can assume $d > 0.064in$ I think):

Table 10–4 Constants A and m of $S_{ut} = A/d^m$ for Estimating Minimum Tensile Strength of Common Spring Wires

Material	ASTM No.	Exponent m	Diameter, In	A, kpsi · In ^m	Diameter, mm	A, MPa · mm ^m	Relative Cost of Wire
Music wire*	A228	0.145	0.004–0.256	201	0.10–6.5	2211	2.6
OQ&T wire†	A229	0.187	0.020–0.500	147	0.5–12.7	1855	1.3
Hard-drawn wire‡	A227	0.190	0.028–0.500	140	0.7–12.7	1783	1.0
Chrome-vanadium wire§	A232	0.168	0.032–0.437	169	0.8–11.1	2005	3.1
Chrome-silicon wire	A401	0.108	0.063–0.375	202	1.6–9.5	1974	4.0
302 Stainless wire#	A313	0.146	0.013–0.10	169	0.3–2.5	1867	7.6–11
		0.263	0.10–0.20	128	2.5–5	2065	
		0.478	0.20–0.40	90	5–10	2911	
Phosphor-bronze wire**	B159	0	0.004–0.022	145	0.1–0.6	1000	8.0
		0.028	0.022–0.075	121	0.6–2	913	
		0.064	0.075–0.30	110	2–7.5	932	

*Surface is smooth, free of defects, and has a bright, lustrous finish.

†Has a slight heat-treating scale which must be removed before plating.

‡Surface is smooth and bright with no visible marks.

§Aircraft-quality tempered wire, can also be obtained annealed.

||Tempered to Rockwell C49, but may be obtained untempered.

#Type 302 stainless steel.

**Temper CA510.

Table 10–5 Mechanical Properties of Some Spring Wires

Material	Elastic Limit, Percent of S_u		Diameter d , in	E		G	
	Tension	Torsion		Mpsi	GPa	Mpsi	GPa
Music wire A228	65–75	45–60	<0.032	29.5	203.4	12.0	82.7
			0.033–0.063	29.0	200	11.85	81.7
			0.064–0.125	28.5	196.5	11.75	81.0
			>0.125	28.0	193	11.6	80.0
HD spring A227	60–70	45–55	<0.032	28.8	198.6	11.7	80.7
			0.033–0.063	28.7	197.9	11.6	80.0
			0.064–0.125	28.6	197.2	11.5	79.3
			>0.125	28.5	196.5	11.4	78.6
Oil tempered A239	85–90	45–50		28.5	196.5	11.2	77.2
Valve spring A230	85–90	50–60		29.5	203.4	11.2	77.2
Chrome-vanadium A231	88–93	65–75		29.5	203.4	11.2	77.2
A232	88–93			29.5	203.4	11.2	77.2
Chrome-silicon A401	85–93	65–75		29.5	203.4	11.2	77.2
Stainless steel							
A313*	65–75	45–55		28	193	10	69.0
17-7PH	75–80	55–60		29.5	208.4	11	75.8
414	65–70	42–55		29	200	11.2	77.2
420	65–75	45–55		29	200	11.2	77.2
431	72–76	50–55		30	206	11.5	79.3
Phosphor-bronze B159	75–80	45–50		15	103.4	6	41.4
Beryllium-copper B197	70	50		17	117.2	6.5	44.8
Inconel alloy X-750	65–70	40–45		31	213.7	11.2	77.2

*Also includes 302, 304, and 316.

- random constants
 $n_s = 1.2$ at solid height as a start, you will check this again later
 $\xi = 0.15$
- get the percentage of S_{sy} you are allowed from this table:

Table 10–6 Maximum Allowable Torsional Stresses for Helical Compression Springs in Static Applications

Material	Maximum Percent of Tensile Strength	
	Before Set Removed (includes K_W or K_B)	After Set Removed (includes K_s)
Music wire and cold-drawn carbon steel	45	60–70
Hardened and tempered carbon and low-alloy steel	50	65–75
Austenitic stainless steels	35	55–65
Nonferrous alloys	35	55–65

- Choose a range of wire diameters (d) from Table A-28 that you think will work:

Table A-28 Decimal Equivalents of Wire and Sheet-Metal Gauges* (All Sizes Are Given in Inches)

Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard†	Manu- facturers Standard	Steel Wire or Washburn & Moen	Music Wire	Stubs Steel Wire	Twist Drill
Principal Use:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lbf/in ²	Ferrous Sheet	Ferrous Wire Except Music Wire	Music Wire	Steel Drill Rod	Twist Drills and Drill Steel
7/0			0.500		0.490			
6/0	0.580 0		0.468 75		0.461 5	0.004		
5/0	0.516 5		0.437 5		0.430 5	0.005		
4/0	0.460 0	0.454	0.406 25		0.393 8	0.006		
3/0	0.409 6	0.425	0.375		0.362 5	0.007		
2/0	0.364 8	0.380	0.343 75		0.331 0	0.008		
0	0.324 9	0.340	0.312 5		0.306 5	0.009		
1	0.289 3	0.300	0.281 25		0.283 0	0.010	0.227	0.228 0
2	0.257 6	0.284	0.265 625		0.262 5	0.011	0.219	0.221 0
3	0.229 4	0.259	0.25	0.239 1	0.243 7	0.012	0.212	0.213 0
4	0.204 3	0.238	0.234 375	0.224 2	0.225 3	0.013	0.207	0.209 0
5	0.181 9	0.220	0.218 75	0.209 2	0.207 0	0.014	0.204	0.205 5
6	0.162 0	0.203	0.203 125	0.194 3	0.192 0	0.016	0.201	0.204 0
7	0.144 3	0.180	0.187 5	0.179 3	0.177 0	0.018	0.199	0.201 0
8	0.128 5	0.165	0.171 875	0.164 4	0.162 0	0.020	0.197	0.199 0
9	0.114 4	0.148	0.156 25	0.149 5	0.148 3	0.022	0.194	0.196 0
10	0.101 9	0.134	0.140 625	0.134 5	0.135 0	0.024	0.191	0.193 5
11	0.090 74	0.120	0.125	0.119 6	0.120 5	0.026	0.188	0.191 0
12	0.080 81	0.109	0.109 357	0.104 6	0.105 5	0.029	0.185	0.189 0
13	0.071 96	0.095	0.093 75	0.089 7	0.091 5	0.031	0.182	0.185 0
14	0.064 08	0.083	0.078 125	0.074 7	0.080 0	0.033	0.180	0.182 0
15	0.057 07	0.072	0.070 312 5	0.067 3	0.072 0	0.035	0.178	0.180 0
16	0.050 82	0.065	0.062 5	0.059 8	0.062 5	0.037	0.175	0.177 0
17	0.045 26	0.058	0.056 25	0.053 8	0.054 0	0.039	0.172	0.173 0

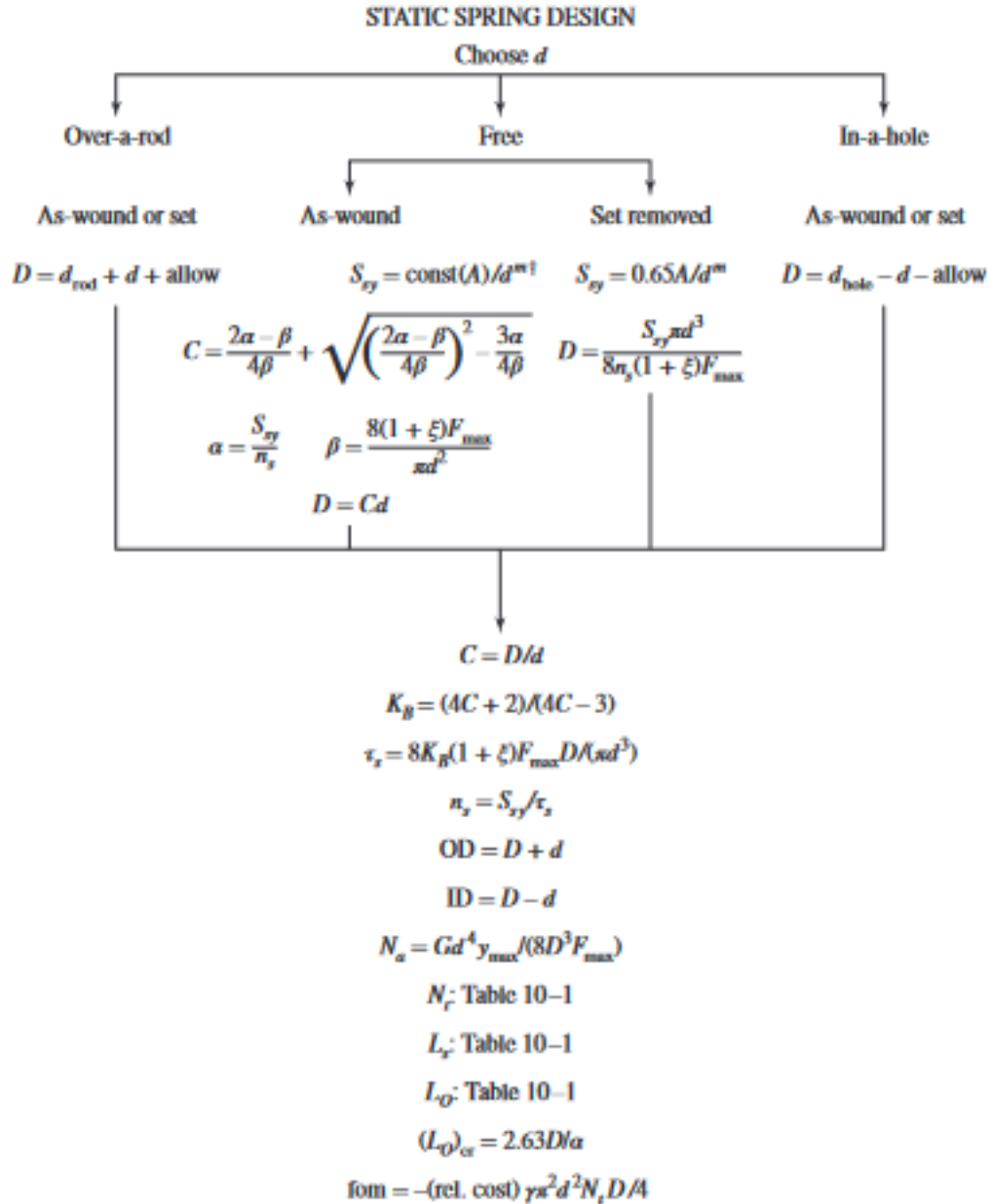
Table A-28 Decimal Equivalents of Wire and Sheet-Metal Gauges* (All Sizes Are Given in Inches) (Continued)

Name of Gauge:	American or Brown & Sharpe	Birmingham or Stubs Iron Wire	United States Standard [†]	Manufacturers Standard	Steel Wire or Washburn & Moen	Music Wire	Stubs Steel Wire	Twist Drill
Principal Use:	Nonferrous Sheet, Wire, and Rod	Tubing, Ferrous Strip, Flat Wire, and Spring Steel	Ferrous Sheet and Plate, 480 lbf/ft ²	Ferrous Sheet	Ferrous Wire Except Music Wire	Music Wire	Steel Drill Rod	Twist Drills and Drill Steel
18	0.040 30	0.049	0.05	0.047 8	0.047 5	0.041	0.168	0.169 5
19	0.035 89	0.042	0.043 75	0.041 8	0.041 0	0.043	0.164	0.166 0
20	0.031 96	0.035	0.037 5	0.035 9	0.034 8	0.045	0.161	0.161 0
21	0.028 46	0.032	0.034 375	0.032 9	0.031 7	0.047	0.157	0.159 0
22	0.025 35	0.028	0.031 25	0.029 9	0.028 6	0.049	0.155	0.157 0
23	0.022 57	0.025	0.028 125	0.026 9	0.025 8	0.051	0.153	0.154 0
24	0.020 10	0.022	0.025	0.023 9	0.023 0	0.055	0.151	0.152 0
25	0.017 90	0.020	0.021 875	0.020 9	0.020 4	0.059	0.148	0.149 5
26	0.015 94	0.018	0.018 75	0.017 9	0.018 1	0.063	0.146	0.147 0
27	0.014 20	0.016	0.017 187 5	0.016 4	0.017 3	0.067	0.143	0.144 0
28	0.012 64	0.014	0.015 625	0.014 9	0.016 2	0.071	0.139	0.140 5
29	0.011 26	0.013	0.014 062 5	0.013 5	0.015 0	0.075	0.134	0.136 0
30	0.010 03	0.012	0.012 5	0.012 0	0.014 0	0.080	0.127	0.128 5
31	0.008 928	0.010	0.010 937 5	0.010 5	0.013 2	0.085	0.120	0.120 0
32	0.007 950	0.009	0.010 156 25	0.009 7	0.012 8	0.090	0.115	0.116 0
33	0.007 080	0.008	0.009 375	0.009 0	0.011 8	0.095	0.112	0.113 0
34	0.006 305	0.007	0.008 593 75	0.008 2	0.010 4		0.110	0.111 0
35	0.005 615	0.005	0.007 812 5	0.007 5	0.009 5		0.108	0.110 0
36	0.005 000	0.004	0.007 031 25	0.006 7	0.009 0		0.106	0.106 5
37	0.004 453		0.006 640 625	0.006 4	0.008 5		0.103	0.104 0
38	0.003 965		0.006 25	0.006 0	0.008 0		0.101	0.101 5
39	0.003 531				0.007 5		0.099	0.099 5
40	0.003 145				0.007 0		0.097	0.098 0

*Specify sheet, wire, and plate by stating the gauge number, the gauge name, and the decimal equivalent in parentheses.

[†]Reflects present average and weights of sheet steel.

- You are going to use this flow chart for every d and decide which one is best (note Free as-wound springs have a different formula for C):



Print or display: $d, D, C, \text{OD}, \text{ID}, N_a, N_f, L_s, L_D, (L_D)_{cr}, n_s, \text{fom}$
 Build a table, conduct design assessment by inspection
 Eliminate infeasible designs by showing active constraints
 Choose among satisfactory designs using the figure of merit
[†]const is found from Table 10-6.

- You should end up with a table very similar to this:

d	0.063	0.067	0.071	0.075	0.080	0.085	0.090	0.095
D	0.391	0.479	0.578	0.688	0.843	1.017	1.211	1.427
C	6.205	7.153	8.143	9.178	10.53	11.96	13.46	15.02
OD	0.454	0.546	0.649	0.763	0.923	1.102	1.301	1.522
N_a	39.1	26.9	19.3	14.2	10.1	7.3	5.4	4.1
L_s	2.587	1.936	1.513	1.219	0.964	0.790	0.668	0.581
I_0	4.887	4.236	3.813	3.519	3.264	3.090	2.968	2.881
$(L_0)_{cr}$	2.06	2.52	3.04	3.62	4.43	5.35	6.37	7.51
n_s	1.2	1.2	1.2	1.2	1.2	1.2	1.2	1.2
fom	-0.409	-0.399	-0.398	-0.404	-0.417	-0.438	-0.467	-0.505

- Now eliminate options based on the conditions:
Design requirements: $4 \leq C \leq 12$, $3 \leq N_a \leq 15$, $\xi \geq 0.15$, $n_s \geq 1.2$ Note n_s is the safety factor at solid height
- Hopefully you only have a few left at most, use the figure of merit (fom) to decide which one is best. Biggest number is best

5.3.4 Calculating based on C (seems better):

- Find an A and m value that fits the needs of the wire for the question, this is mostly material dependent. Note W&M wire is hand drawn:

Table 10–4 Constants A and m of $S_{ut} = A/d^m$ for Estimating Minimum Tensile Strength of Common Spring Wires

Material	ASTM No.	Exponent m	Diameter, In	A , kpsi $\cdot$ In ^{m}	Diameter, mm	A , MPa $\cdot$ mm ^{m}	Relative Cost of Wire
Music wire*	A228	0.145	0.004–0.256	201	0.10–6.5	2211	2.6
OQ&T wire†	A229	0.187	0.020–0.500	147	0.5–12.7	1855	1.3
Hard-drawn wire‡	A227	0.190	0.028–0.500	140	0.7–12.7	1783	1.0
Chrome-vanadium wire§	A232	0.168	0.032–0.437	169	0.8–11.1	2005	3.1
Chrome-silicon wire	A401	0.108	0.063–0.375	202	1.6–9.5	1974	4.0
302 Stainless wire#	A313	0.146	0.013–0.10	169	0.3–2.5	1867	7.6–11
		0.263	0.10–0.20	128	2.5–5	2065	
		0.478	0.20–0.40	90	5–10	2911	
Phosphor-bronze wire**	B159	0	0.004–0.022	145	0.1–0.6	1000	8.0
		0.028	0.022–0.075	121	0.6–2	913	
		0.064	0.075–0.30	110	2–7.5	932	

*Surface is smooth, free of defects, and has a bright, lustrous finish.

†Has a slight heat-treating scale which must be removed before plating.

‡Surface is smooth and bright with no visible marks.

§Aircraft-quality tempered wire, can also be obtained annealed.

||Tempered to Rockwell C49, but may be obtained untempered.

#Type 302 stainless steel.

**Temper CA510.

- Use this table to get a percentage of S_{ut} you are allowed

Table 10–6 Maximum Allowable Torsional Stresses for Helical Compression Springs in Static Applications

Material	Maximum Percent of Tensile Strength	
	Before Set Removed (includes K_W or K_B)	After Set Removed (includes K_s)
Music wire and cold-drawn carbon steel	45	60–70
Hardened and tempered carbon and low-alloy steel	50	65–75
Austenitic stainless steels	35	55–65
Nonferrous alloys	35	55–65

- find S_{sy}
 $S_{sy} = \text{percent from above} \cdot S_{ut} = \text{percent} \cdot \frac{A}{d^m}$

- Solve for d using $C = 10$ since we just guess C and the example used 10:

$$K_B = \frac{4C+2}{4C-3}$$

$$d = (0.163K_B \frac{C}{A})^{1/(2-m)}$$

- Use the wire that most closely fits what you calculated from table A-28:

- check safety factor:

$$n_s = 7.363 \frac{Ad^{2-m}}{K_B C}$$

If it is way over or way under 1.2 you can try a different d close to what you calculated, or don't they'll probably mark it right either way

- Find the spring rate k :

$$k = \frac{F}{y}$$

- Find the number of active coils:

$$N_a = \frac{dG}{8kC^3}$$

if this is outside the condition $3 \leq N_a \leq 15$ adjust C while keeping it within $4 \leq C \leq 12$, if you can't do this you're fuck try another d maybe idk

- Find number of coils (N_t) from this table:

Table 10–1 Formulas for the Dimensional Characteristics of Compression Springs (N_a = Number of Active Coils)

Term	Type of Spring Ends			
	Plain	Plain and Ground	Squared or Closed	Squared and Ground
End coils, N_e	0	1	2	2
Total coils, N_t	N_a	$N_a + 1$	$N_a + 2$	$N_a + 2$
Free length, L_0	$pN_a + d$	$p(N_a + 1)$	$pN_a + 3d$	$pN_a + 2d$
Solid length, L_s	$d(N_t + 1)$	dN_t	$d(N_t + 1)$	dN_t
Pitch, p	$(L_0 - d)/N_a$	$L_0/(N_a + 1)$	$(L_0 - 3d)/N_a$	$(L_0 - 2d)/N_a$

- Calculate deflection length:

$$y_s = \frac{F_{max}}{k}$$

- Use Table 1 above to solid length, free length and then solve for mean coil diameter:

$$D = Cd$$

$OD = D + d$ Remember some of these forms may be different for different boundary type of

springs, refer to the flow chart above to confirm formulas

- Calculate the alpha condition:

$$\alpha < 2.63 \frac{D}{L_0}$$

- Use this table to see how you can support the spring:

Table 10–2 End-Condition Constants α for Helical Compression Springs*

End Condition	Constant α
Spring supported between flat parallel surfaces (fixed ends)	0.5
One end supported by flat surface perpendicular to spring axis (fixed); other end pivoted (hinged)	0.707
Both ends pivoted (hinged)	1
One end clamped; other end free	2

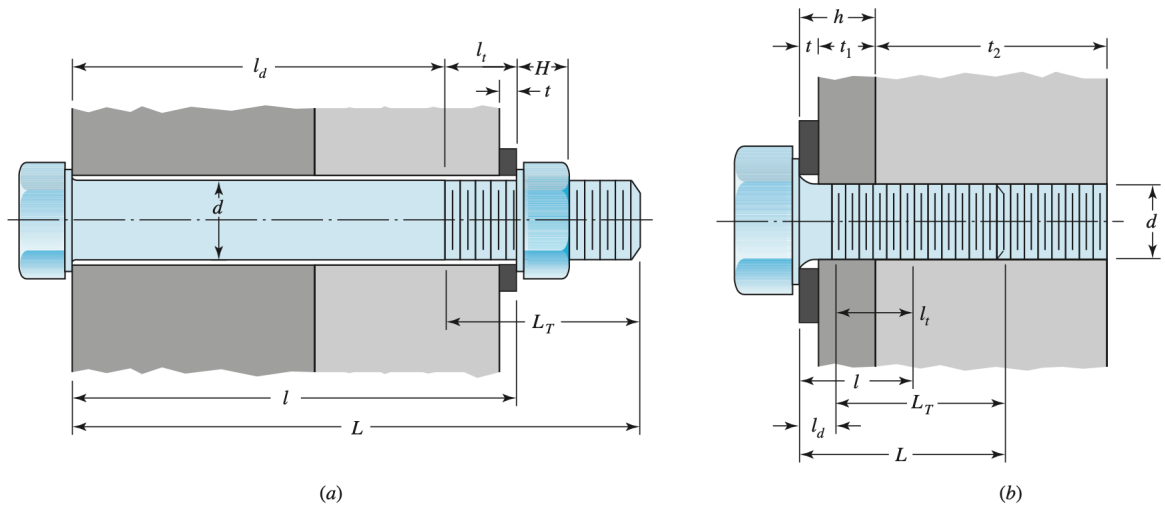
*Ends supported by flat surfaces must be squared and ground.

5.4 Fasteners/Bolts

5.4.1 Design Selection

1. Determine a suitable length for the bolt:

- (a) Compute the grip length.



- If your bolt arrangement is like figure (a):

l = thickness of all material squeezed between face of bolt and face of nut